Capstone Three — Project Proposal

Optimizing Term Deposit Marketing Campaigns for a European Bank

Colin Segura | Springboard Data Science Career Track | Business Insights Specialization

Problem statement

A Portuguese retail bank ran a marketing campaign over the phone, trying to sign existing customers up for term deposits. Roughly eleven of every hundred calls actually led to a subscription; the other eighty nine still cost the bank real money in agent time, telephony, and operational overhead. What I want to work out is whether we can look at the campaign's history, combined with what the bank already knows about each customer before it dials, and predict with some confidence who is likely to say yes. If the answer is yes, the next campaign spends fewer calls chasing better prospects and returns more revenue per dollar spent.

Context

Direct marketing sits near the top of the recurring cost list in retail banking, and most banks still lean on hand tuned segmentation rules to decide who to call. When the underlying conversion rate is only around eleven percent, even modest gains in targeting move ROI substantially — the same revenue for fewer calls, or meaningfully more revenue for the same call volume. I chose this problem partly because it mirrors the client analytics work my team at JPMorganChase does for wealth management outreach. Working through it on public data gives me a portfolio version of a decision I have watched real product and marketing teams make.

Criteria for success

Success has two layers for me. The first is quantitative: the final model needs to clearly beat two baselines on held out data — a majority class predictor, and a simple logistic regression on the three most intuitive features — measured by ROC AUC, precision at the top decile of predicted probabilities (the number marketers actually track), and expected profit at the model's best threshold given reasonable assumptions about cost per call and revenue per subscription. The second layer is that the model has to be usable. I want the report to include a lift chart and a threshold versus profit chart clear enough that a non technical marketing lead could read them, decide where to draw the outreach cutoff, and defend the choice in a meeting.

Scope of solution space

I plan to use all 41,188 client records in the UCI Bank Marketing dataset, which cover 45 months of campaign activity from May 2008 through November 2010. The target is binary: did the customer subscribe. The modeling side starts with logistic regression as an interpretable baseline and moves to gradient boosted trees (XGBoost and LightGBM) as the primary models, with SHAP values for interpretation. A few things sit deliberately outside the scope. Call duration gets dropped because it perfectly leaks the outcome, since a call of zero seconds is by definition a non subscription. The deliverable is a batch scoring model, not a real time inference service. And while I will use macroeconomic features like the Euribor rate wherever they add signal, I am not trying to generalize beyond the Portuguese context or write a paper about which macro variables matter and why.

Constraints

A few honest constraints. The data covers 2008 through 2010, which spans the acute phase of the financial crisis, but it is now more than fifteen years old — so findings about specific customer segments should be treated as methodological demonstrations rather than statements about the current market. Class imbalance runs at about eleven percent positives, which rules out plain accuracy as a metric and may require class weight handling during fitting. The dataset also carries no real customer financial history — no account balances, no transaction data, no full CRM view — which caps the model's ceiling below what a bank could achieve using its own internal data. And the time budget is roughly 80 to 120 hours across wrangling, EDA, modeling, and reporting.

Stakeholders

The model and its findings would land in front of five audiences. Marketing analytics owns the targeting logic and is the primary consumer of the ranked prospect list. Call center operations uses that list to prioritize agent time and size staffing for the next campaign. The term deposit product team gets qualitative signal about which customer attributes track with subscription. Finance uses the projected lift for revenue forecasting and marketing ROI attribution. Executive leadership sees the summary charts when weighing channel mix decisions across direct marketing, digital, and branch. There is also a sixth audience worth naming: a hiring manager reviewing me for a marketing or customer analytics role could read the final report and get a real sense of how I approach a business decision, not just a modeling exercise.

Data sources

- UCI Bank Marketing dataset — 41,188 client records across 20 input columns plus the binary target. Client attributes include age, occupation, marital status, education, and credit or loan status. Campaign attributes include contact channel, month and day of last contact, days since prior contact, and prior campaign outcome. Macroeconomic attributes include employment variation rate, consumer price index, consumer confidence index, and Euribor 3 month rate. This is the primary data source.
- Kaggle mirror — Same file as the UCI source but easier to download directly from a Kaggle-hosted CSV.
- Reference paper (Moro, Cortez & Rita, 2014) — The original 'A Data Driven Approach to Predict the Success of Bank Telemarketing' publication in Decision Support Systems, cited on the UCI page. Gives the business context and prior modeling baselines to compare against.

Approach

The work follows the same Data Science Method sequence I used for Capstone Two. Wrangling takes care of the categorical encodings, the unknown value sentinels, and the deliberate drop of the leaky call duration column. EDA covers univariate distributions, subscription rate by feature bin, and correlations with the target. Preprocessing sets up an 80/20 stratified train/test split with imputation and one hot encoding wrapped inside a scikit-learn Pipeline. Modeling begins with logistic regression and moves to XGBoost and LightGBM with cross validated hyperparameter tuning. SHAP values handle interpretation. The final deliverables are a GitHub repository, a project report, a slide deck, and a short targeting decision memo written for the marketing analytics stakeholder.